TOPIC 5.5

Regularization

Ridge (L2), Lasso (L1) and Elastic Net: keeping coefficients under control

WHAT YOU WILL LEARN

- Explain **why** regularization is needed and what it does to a model
- Write the **Ridge, Lasso and Elastic Net** objectives and say what each penalty does
- Describe the effect of the strength **α** (alpha)
- Know why **Lasso can remove features** while Ridge only shrinks them
- Choose between the three, and know why features must be **scaled** first

1. The problem: coefficients that grow too large

An ordinary linear regression picks coefficients that make the training error as small as possible. When features overlap (are correlated) or there are many features compared with the number of rows, it may use huge positive and negative coefficients that cancel each other out. The model then fits the training data very closely but is **unstable and overfits**: a small change in the data changes the coefficients a lot (this is the high variance of note 5.4).

DEFINITION **Regularization**

Adding a **penalty for large coefficients** to the quantity the model minimises. The model must now balance two aims: **fit the data well** and **keep the coefficients small**.

$$\text{Loss} = \sum_i (y_i - \hat{y}_i)^2 \; + \; \alpha \times \text{Penalty}(w)$$

The number **α** (alpha) sets how strict the penalty is. With **α = 0** there is no penalty and we get ordinary regression. A larger α pushes the coefficients closer to zero, giving a simpler model (more bias, less variance).

2. Ridge regression (L2 penalty)

Ridge adds the **sum of the squared coefficients**:

$$\min_w \; \sum_i (y_i - \hat{y}_i)^2 + \alpha \sum_j w_j^2$$

- Large coefficients are punished heavily (because of the square), so they all get **pulled towards zero**.
- Coefficients become **small but not exactly zero**: Ridge keeps every feature.
- Works well when many features each contribute a little, and when features are correlated: it spreads the weight among them.

3. Lasso regression (L1 penalty)

Lasso adds the **sum of the absolute values** of the coefficients. (Lasso stands for *Least Absolute Shrinkage and Selection Operator*.) scikit-learn writes its objective as:

$$\min_w \ \frac{1}{2n}\sum_i (y_i - \hat{y}_i)^2 + \alpha \sum_j |w_j|$$

- The absolute-value penalty can push weak coefficients to **exactly zero**, which removes those features from the model.
- So Lasso does **feature selection** automatically (this is the embedded method mentioned in note 5.2).
- With a group of correlated features, Lasso tends to keep one and drop the others.

4. Elastic Net (L1 and L2 together)

Elastic Net mixes both penalties. In scikit-learn, **l1_ratio** (written ρ here) says how much of each:

$$\min_w \ \frac{1}{2n}\sum_i (y_i - \hat{y}_i)^2 + \alpha\rho \sum_j |w_j| + \frac{\alpha(1-\rho)}{2}\sum_j w_j^2$$

With ρ = 1 it is Lasso; with ρ = 0 it is Ridge. It keeps Lasso's ability to remove features while being steadier than Lasso when features are correlated.

> **NOTE** **A note on α in code**
> Different libraries scale the squared-error term differently (Ridge uses the plain sum, while Lasso and Elastic Net divide by 2n). So the same number for α does **not** mean the same strength in Ridge and Lasso. Compare models by their results, not by α alone.

	Ridge	Lasso	Elastic Net
Penalty	Σ w² (L2)	Σ |w| (L1)	Mix of both
Coefficients become	Small, never exactly 0	Some exactly 0	Some exactly 0
Selects features?	No	Yes	Yes
Correlated features	Shares the weight	Keeps one, drops others	Handles groups better
scikit-learn	Ridge	Lasso	ElasticNet

5. A small worked example (one feature)

Take five points with feature x = −2, −1, 0, 1, 2 and y = −3.1, −1.4, 0.2, 1.6, 3.0 (after subtracting the mean of y). Here Σxy = 15.2 and Σx² = 10, so ordinary regression gives slope = 15.2 / 10 = **1.52**.

For one centred feature the penalised slopes have simple formulas, which show the difference between the two penalties:

- **Ridge:** slope = Σxy / (Σx² + α). The slope **shrinks smoothly** towards zero but never reaches it.
- **Lasso:** slope = (|Σxy/n| − α) / (Σx²/n), floored at 0, with the sign of Σxy. Once α is large enough the slope becomes **exactly 0**.

Ridge α	Ridge slope	Lasso α	Lasso slope
0	1.520	0	1.520
2	1.267	1	1.020
10	0.760	2	0.520
50	0.253	3	0.020
		4	0.000

Ridge shrinks the slope step by step (0.25 even at α = 50) but never reaches zero. Lasso takes the slope down in equal steps and hits **exactly 0** at α = 3.04 (= |Σxy/n|). This is why Lasso removes features and Ridge does not.

6. Why Lasso gives exact zeros: the geometry

Regularization can be pictured as a **budget** on the coefficients. We look for the best fit that stays inside a shaped region around zero. The grey rings are lines of equal training error, centred on the unpenalised best fit.

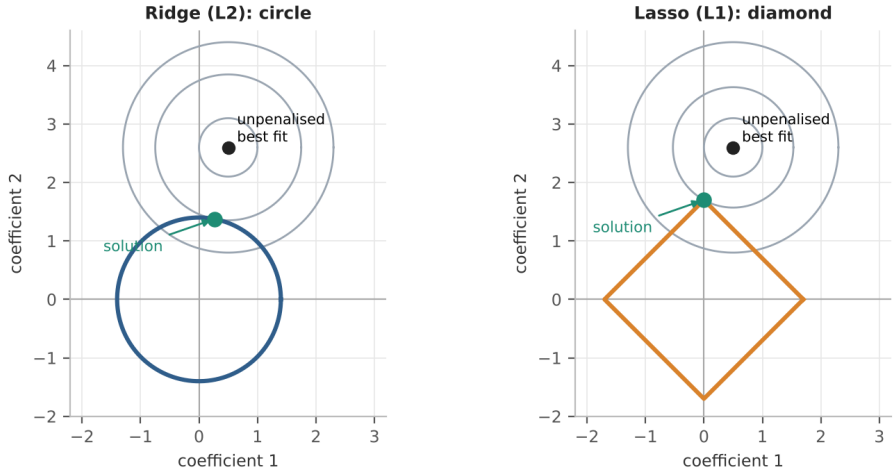


Figure 1. Ridge's region is a circle; Lasso's is a diamond with corners on the axes. The best fit inside the region often lands on a Lasso corner, where one coefficient is exactly 0 (in the picture, coefficient 1). Ridge's circle has no corners, so neither coefficient is exactly 0.

7. The effect of α on real data

The class data set again: predict `mpg` (fuel economy) from the 10 car features (weights, engine, year, origin and so on). Figure 2 follows every coefficient as α grows.

- **Small α:** almost no penalty, results close to ordinary regression (possible overfitting).
- **Large α:** coefficients are crushed towards zero (underfitting: too simple).
- **In between:** the sweet spot, found by **cross-validation** (see note 5.3). scikit-learn offers `RidgeCV`, `LassoCV` and `ElasticNetCV` for this.

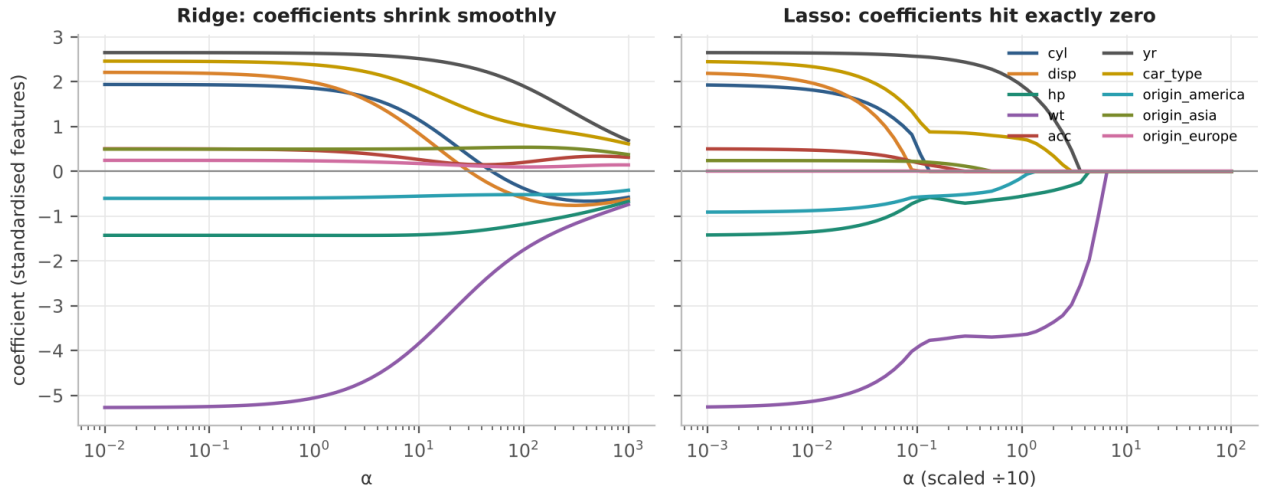


Figure 2. Coefficient paths. Left: Ridge shrinks all coefficients towards zero. Right: Lasso sets them to exactly zero one by one, weakest first.

8. Comparing the models

Model	Train R^2	Test R^2	Coefficients ≠ 0
Linear (no penalty)	0.8362	0.8439	10 of 10
Ridge ($\alpha = 10$)	0.8306	0.8360	10 of 10
Lasso ($\alpha = 0.5$)	0.8109	0.8224	6 of 10
Elastic Net ($\alpha = 0.5$, $\rho = 0.5$)	0.7966	0.8046	9 of 10

In this data set, ordinary regression is already fairly stable, so the penalties lower test accuracy only a little (R^2 stays between 0.80 and 0.84). What changes is the **model**: Ridge makes the coefficients smaller, and Lasso with $\alpha = 0.5$ keeps 6 of 10 features, giving a simpler model with slightly lower accuracy. Regularization pays off most when there are many features, strong overlap between them, or little data.

> **IMPORTANT** **Scale your features first**
>
> The penalty adds up the sizes of the coefficients, and a coefficient's size depends on the units of its feature. A feature measured in thousands gets a tiny coefficient and is barely penalised; one measured in fractions gets a huge coefficient and is penalised hard. **Standardise the features** (see note 5.6) before using Ridge, Lasso or Elastic Net, and fit the scaler on the **training rows only**. (Here that was done properly. In the class notebook the whole data set was scaled before splitting, so its numbers differ slightly from these.)

9. Which one should I use?

- Many features that all matter a little, correlated features: **Ridge**.
- You want a **smaller model** and suspect many features are useless: **Lasso**.
- Many correlated features **and** you want some removed: **Elastic Net**.
- Not sure: try all three with cross-validation and compare test scores.

Summary

SUMMARY

- Regularization adds a **penalty on large coefficients** to reduce overfitting (it trades a little bias for a lot less variance)
- **Ridge (L2)**: penalty Σw^2. Shrinks all coefficients, never exactly zero
- **Lasso (L1)**: penalty $\Sigma |w|$. Can set coefficients to **exactly zero**, so it selects features
- **Elastic Net**: mix of both, controlled by l1_ratio ρ
- **α** controls strength: $\alpha = 0$ is ordinary regression; very large α gives an over-simple model. Choose it by cross-validation
- **Scale features first**, fitting the scaler on training data only

Practice questions

Try each question before reading the answer underneath it.

Q1. What does the penalty in regularization do?

Answer. It punishes large coefficients, so the model must balance fitting the data against keeping its coefficients small. This makes the model simpler and less prone to overfitting.

Q2. What is the difference between Ridge and Lasso?

Answer. Ridge penalises the **squares** of the coefficients and only shrinks them. Lasso penalises their **absolute values** and can make some exactly zero, which removes those features.

Q3. What happens as α increases?

Answer. The coefficients shrink further, and the model becomes simpler. Too large an α causes underfitting. At $\alpha = 0$ we get ordinary regression.

Q4. When would you choose Elastic Net?

Answer. When features are correlated and you also want some of them removed: it combines Ridge's stability with Lasso's feature selection.

Q5. Why must features be scaled before regularization?

Answer. The penalty depends on coefficient size, and coefficient size depends on the feature's units. Without scaling, features measured in large units are penalised too little and those in small units too much.

Q6. In Lasso a coefficient becomes exactly 0. What does that mean for the feature?

Answer. The model no longer uses that feature at all: Lasso has selected it out.